

Abstract No.

Characterization of ZBTB18 C-terminal short form function in glioblastoma

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Zinc finger and BTB domain-containing protein 18 (ZBTB18) is a key player in both normal nervous system development and cancer, where it behaves as a tumor suppressor and negative regulator of glioblastoma (GBM) mesenchymal genes. ZBTB18 is cleaved by the calpain 2 (CAPN2) protease, which generates two functionally different fragments (SF-Nte and SF-Cte), SF-Nte acquiring oncogenic functions. However, whether the SF-Cte still retains ZBTB18 tumor suppression function needs to be determined.

Here, we have characterized the regulation of ZBTB18 SF-Cte and investigated its function. Our preliminary data suggest that ZBTB18 SF-Cte levels are low due to protein instability. Interestingly, RNAseq data suggest that ZBTB18 SF-Cte retains, at least partly, ZBTB18 suppressive function in tumor. Whether new protein interactions and/or cellular activities are gained is a matter of investigation. Attempts to decipher ZBTB18 SF-Cte interactome suggested the interaction with proteins involved in its degradation. Further investigation of ZBTB18 SF-Cte function and regulation will be important to understand whether targeting key ZBTB18 SF-Cte regulators to increase its stability, could be a therapeutic option to treat glioblastoma.



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